Customer Churn & Retention Prediction System

SYNKOC AI & ML Pro Capstone Project

Dataset: Telco Customer Churn. Raw rows: 7,043. Clean modeling rows: 7,032. Churned customers: 1,869.

Model Evaluation

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Logistic Regression	0.726	0.490	0.797	0.607	0.835	0.618
Random Forest	0.768	0.548	0.717	0.621	0.831	0.633

Business Outputs

High-risk customers (probability >= 60%): 1,907. Estimated aggregate monthly revenue exposure based on probability-weighted monthly charges: ■185,270.27 (dataset currency units are not guaranteed to be INR).

The report documents offline model evaluation and business-oriented risk scoring. Predictions are not guarantees of individual customer behavior.